

Technical Documentation & Developer Reference Manual

Codebase Architecture, Dependency Management, Model Retraining Regimes & Environment Setup

Academic and strategic research artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2). Demonstrating circular economy transition, machine learning churn mitigation, FunkSVD recommendation, and localized generative AI edge deployment.

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1. Codebase Directory Map

```
SDC2/
├── backend/                # Pure MongoDB Django REST Gateway
│   ├── core/              # settings.py (DATABASES={}), views_chat.py, urls.py
│   ├── appliances/        # MongoEngine Appliance catalog models
│   ├── bookings/          # Rental state machine & agreements
│   ├── ml_churn/           # LightGBM pipeline & model_lightgbm.pkl
│   ├── ml_recommend/      # FunkSVD matrix factorization & model.pkl
│   ├── tests/             # Automated verification: test_all_endpoints.py
│   └── requirements.txt    # Python dependencies
├── frontend/              # React 19 + Vite Storefront & Admin
│   ├── src/components/    # AIChatbot.jsx, Navbar, Catalog
│   └── package.json       # Node dependencies
├── scripts/               # Automated orchestration & seeding
│   ├── start_all.bat      # 1-Click launcher (Mongo, Ollama, Django, Vite)
│   └── seeding/           # Catalog populators
└── docs/pdf_deliverables/ # 20 Academic & Technical PDF Deliverables
```

2. Environment Setup & Dependency Installation

2.1 Backend Setup

```
cd backend
python -m pip install -r requirements.txt
```

2.2 Frontend Setup

```
cd frontend
npm install
npm run dev
```

3. Machine Learning Model Retraining Commands

To refresh the machine learning pipelines against new empirical interactions:

```
# Retrain LightGBM Churn Pipeline with 5-Fold Stratified CV
python manage.py train_churn

# Retrain Biased FunkSVD Collaborative Filtering Recommender
python manage.py train_recommender

# Retrain Multi-Category ARIMA Demand Forecasting
python manage.py train_forecast
```

4. Automated One-Click System Launcher

Run `scripts/start_all.bat` (or `start_all.ps1`). The script sequentially verifies and launches MongoDB on port 27017, Ollama on port 11434, Django REST on port 8000, and Vite Frontend on port 5173, subsequently launching the default web browser.